Jayasankar R

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Summary

Embedded Systems and Software Engineer with a Bachelor's degree in Electrical and Electronics Engineering and 2+ years of software engineering experience. Strong foundation in C programming, microcontrollers, control systems, embedded software concepts, and hardware-software integration. Hands-on experience with Arduino-based systems, power electronics, and MATLAB/Simulink modeling. Interested in low-level software development including Flash drivers, bootloaders, device drivers, and automotive embedded systems.

Skills

Programming Languages: C, Embedded C, Java, TypeScript, JavaScript

Embedded Systems: Microcontrollers, Interrupts, Timers, ADC, PWM, GPIO, Flash Memory Fundamentals, Bootloader Concepts

Communication Protocols: UART, I2C, SPI, CAN

Control Systems: PI Control, PID Control, Fuzzy Logic, System Modeling

Tools: MATLAB, Simulink, Git, GitHub Actions, Jenkins

Hardware: Arduino, ESP8266, Sensors, Power Electronics, Solar Energy Systems

Software Development: Spring Boot, Node.js, PostgreSQL, AWS, Docker, Kubernetes

Experience

Onfuturesites Aug 2025 – Feb 2026

Software Engineer

- Worked closely with cross-functional teams to build scalable backend APIs using Node.js and Fastify, supporting thousands of daily requests with high reliability
- Built user-facing features using Next.js and React Native, improving usability and customer engagement
- Architected PostgreSQL schemas handling large datasets (100k+ records), improving query efficiency and data consistency
- Enhanced backend performance by optimising queries and service logic, reducing API latency by 30%
- Contributed to architectural decisions in a fast-paced startup environment, ensuring scalability and maintainability

Technologies: Node.js, Fastify, React Native, Next.js, PostgreSQL, Prisma

DevOps Engineer | Freelance June 2025 – Oct 2025

- Built and executed CI/CD pipelines using GitHub Actions and ArgoCD, reducing deployment time by 60% and enabling near-zero manual intervention
- Containerised and deployed microservices on Kubernetes (AWS EKS), improving scalability and deployment consistency
- Partnered with development teams to improve deployment workflows
- Implemented monitoring and alerting systems using Prometheus and Grafana, reducing issue detection time and improving system reliability
- Integrated development and deployment workflows, reducing manual effort and improving deployment

efficiency across teams

Technologies: ArgoCD, GitHub Actions, Kubernetes (EKS), Helm, Prometheus, Grafana, Loki, Fluent Bit, Docker, AWS (ECR)

Captain Fresh Aug 2023 – April 2025

Full-Stack Engineer

- Engineered backend services for ERP systems handling high-volume stock transfer workflows, improving operational throughput by 15%
- Built scalable REST APIs using TypeScript and PostgreSQL for complex business operations with high-frequency transactions
- Optimised PostgreSQL queries (joins, indexing), reducing API response time by 150–200ms
- Built Spring Boot microservices enabling secure enterprise integrations, improving system security and reliability
- Engineered asynchronous communication between ERP stock transfer and logistics microservices using Kafka, improving system decoupling and enabling reliable event-driven processing
- Applied caching strategies, reducing API response times by 14% and decreasing database load significantly
- Partnered with product and QA teams in Agile sprints, delivering 15+ features and improving release efficiency and time-to-market

Technologies: TypeScript, React.js, PostgreSQL, Sequelize, Spring Boot, Golang, Node.js, AWS

Projects

Heat Exchanger Control System Using Fuzzy Logic Nov 2025 – Dec 2025

- Developed a mathematical model of a plate heat exchanger to analyze transient thermal behavior
- Designed and implemented PI, PID, and Fuzzy Logic controllers using MATLAB/Simulink
- Simulated varying flow-rate and temperature conditions to evaluate controller performance
- Analyzed settling time, overshoot, and steady-state error using generated response curves
- Demonstrated improved stability and reduced settling time using Fuzzy Logic control compared to conventional PID control

Technologies: MATLAB, Simulink, Control Systems, Numerical Modeling

Smart Hybrid Inverter Jul 2022 – Dec 2022

- Designed and developed an embedded control system integrating solar, battery, and grid power sources
- Implemented Incremental Conductance MPPT algorithm for maximum power extraction
- Developed firmware using Arduino and ESP8266 for battery monitoring and automatic source switching
- Interfaced voltage and current sensors using ADC peripherals for real-time measurements
- Implemented automated power-source selection based on battery and load conditions
- Worked with power electronics components including MOSFET switching circuits and DC-DC converters

Technologies: Embedded C, Arduino, ESP8266, MPPT, ADC, Power Electronics

UART-Based Flash Bootloader Jun 2026

- Developed a custom bootloader in Embedded C for firmware update applications
- Implemented UART-based communication for firmware transfer
- Added Flash erase, write, and verification routines
- Designed firmware integrity validation using checksum verification
- Implemented boot sequence logic for safe transition to application firmware

Technologies: Embedded C, UART, Flash Memory, Bootloader Design

Education

Govt. Model Engineering College, Bachelor of Technology in Electrical and Electronics Engineering — Aug 2019 – June 2023